

BINF6999 Poster Brainstorm

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Notes on how the poster was put together, covering the three expert-level criteria in the specification rubric: core message, creativity and impact, and walk-away value. I built two full versions rather than sketching alternatives, so the comparison below is between two things that actually exist.

1. Core message

The one line I want someone to leave with:

Fibrosis is the default program in injured tendon. Innervation is what suppresses it — and that program can be measured with 23 genes.

I went in expecting the opposite. The working hypothesis was that nerve signalling switches a repair program on, and that we would find it. We didn't. Fifteen fibrotic transcription-factor programs turned up in both datasets and not one regenerative program did. If that holds, the therapeutic target is suppressing scar rather than stimulating repair, which is a genuinely different way to think about treating these injuries.

That reversal is the most interesting thing I have, so the poster is built around it rather than around the pipeline.

Quick test for the layout: cover everything except the title and the regulon scatter. If the point doesn't land in about five seconds from three metres back, the layout isn't doing its job.

2. Two versions built

Plan A — three-act narrative. Three numbered columns, each headed by a full sentence instead of a section label, reading left to right: the two cell types and the decision they face, the search for a repair switch and the failure to find one, then the score built from what was actually there. Underneath, a full-width band listing the claims that were stress-tested, and a closing walk-away statement beside the limitations.

Plan B — standard four-section. Introduction, Methods, Results and Conclusions, with Results given two columns because that is where the figures are. Conclusions runs full-width across the bottom. Methods gets a proper column of its own: dataset cards, the six-step pipeline, and the robustness checks.

	Plan A — narrative	Plan B — four-section
Title	States the finding as a claim	Formal project title
Structure	Three acts, full-sentence headers	Introduction / Methods / Results / Conclusions
Methods	Folded into the act openers	Its own column, fully spelled out
The null result	The centrepiece — act 2 is built on it	One bullet among three in Conclusions
Robustness checks	Dedicated full-width band	Inside Methods, where reviewers look for them

Strength	Memorable, and the reversal has room to land	Familiar, easy to follow, complete on method
Risk	Unconventional if the room is formal	Lower ceiling on originality

The content is identical in both. Only the structure and the title differ.

I would go with Plan A. The finding is a reversal, and three acts give it room to land in a way that a Results column doesn't — in Plan B the same sentence ends up as one bullet among three and loses its force. Plan B is the safer choice if the audience skews conservative, or if I would rather field questions about method than about framing.

3. Creativity and impact

Most of the posters I looked at either hide a null result or leave it out entirely. Mine is built on one. That felt risky at first, but it is the more interesting story and it is certainly the more honest one.

A few specific choices:

The regulon scatter shows the zero as an empty quadrant rather than a bar of height zero. You can see that nothing is there, which is hard to convey with a bar chart.

Teal always means regenerative, TSPC, innervated; red always means fibrotic, T-FAP, denervated. Same in all seven figures, so nobody has to go back to a legend to read the next panel.

The retractions are on the poster. Two of three headline claims didn't survive proper re-testing and I say so directly, with the test and the p-value. It is the best answer I have to "how do you know any of this is real?", and I would rather raise it myself than be asked.

4. Walk-away value

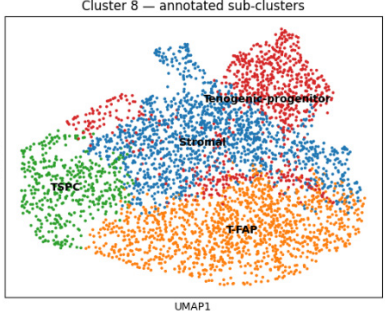
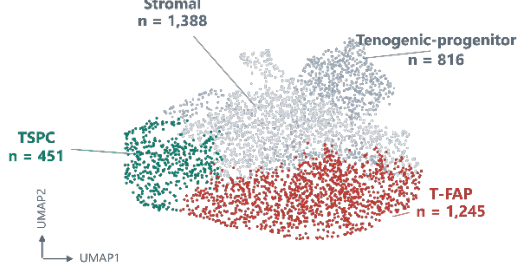
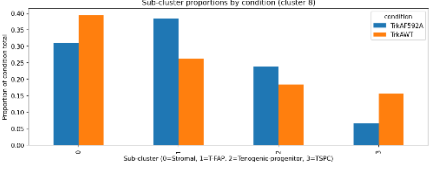
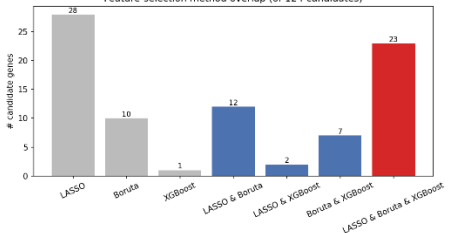
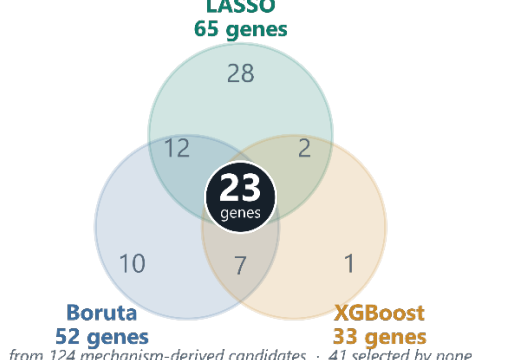
What someone actually leaves with depends on how long they stop:

Five seconds , walking past	Tendon scars by default; nerves hold it back. This has to come from the title and the fifteen-versus-zero contrast alone.
Thirty seconds , stopping	Fifteen fibrotic regulons replicate across two independent datasets and zero regenerative ones do. A 23-gene score built from that program separates the two fates in held-out data at 0.989 ROC-AUC and tracks innervation status.
Three minutes , in conversation	The data redirected the hypothesis. Two headline findings were retracted under testing. Neither dataset has biological replicates, so the real test is new lab data.

The point to make out loud: if fibrosis is the default rather than the failure mode, then treatment should aim at suppression rather than stimulation, and the Healing Index gives a number for whether that is working.

5. Figures

Every figure was rebuilt for the poster rather than lifted from the analysis notebooks. The notebook versions were fine for working but wrong for a wall — titles duplicating the caption, legends where a direct label would do, and type sized for a laptop.

What changed	Before	After
Sub-cluster UMAP. The labels sat on the data and overlapped it, and the colours were whatever matplotlib picked. Now each population is labelled outside the cloud with a leader line and its n, and the two competing fates are the only ones in colour.		
Composition shift. Grouped bars made you subtract to see the change, and the sub-clusters were numbered 0–3 with a decoder key buried in the axis label. As a slope chart the change <i>is</i> the slope, the populations are named, and the conditions are in plain language instead of genotype codes.		
Feature selection. A bar chart of all seven intersections technically contained the answer but made you assemble it yourself. As a Venn the number that matters sits in the middle, and three methods agreeing is obvious at a glance.		

Across all seven figures the same three things changed: titles came off, since the caption already says what the panel is; legends became direct labels wherever possible; and the backgrounds went transparent so the figures sit on the poster rather than in white boxes.